

Program- 7

Implement K-means, K-modes clustering to find natural pattern in data

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

# Numerical data: Height and Weight
data_num = {
    'Height': [150, 160, 170, 180, 190, 200, 210],
    'Weight': [50, 60, 70, 80, 90, 100, 110]
}

df_num = pd.DataFrame(data_num)

# Apply K-Means
kmeans = KMeans(n_clusters=2, random_state=0)
df_num['Cluster'] = kmeans.fit_predict(df_num)

# Output Result
print("=== K-MEANS Clustering Result ===")
print(df_num)

# Visualize
plt.scatter(df_num['Height'], df_num['Weight'], c=df_num['Cluster'], cmap='viridis')
plt.xlabel('Height')
plt.ylabel('Weight')
plt.title('K-Means Clustering (Height vs Weight)')
plt.grid(True)
plt.show()
```

output-

```
=== K-MEANS Clustering Result ===
   Height  Weight  Cluster
0     150     50        0
1     160     60        0
2     170     70        0
3     180     80        0
4     190     90        1
5     200    100        1
```

6 210 110 1

